Discriminant and Logit Analysis

Chapter 18

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Learning Objectives

After this chapter you should be able to:

1. Describe the concept of **discriminant analysis**, its objectives and its applications in marketing research.
2. Outline the procedure: formulation, estimation of the discriminant function, significance, interpretation and validation.
3. Distinguish **two-group** from **multiple** discriminant analysis.
4. Quantify classification accuracy with a **confusion matrix**, and judge a model by its different error rates (false positives vs. false negatives).
5. Explain stepwise discriminant analysis and the Mahalanobis procedure.
6. Describe the **binary logit** model and its advantages over discriminant and regression analysis.
7. Run both techniques in SPSS and SAS.

Agenda

Basic Concept
The Discriminant Analysis Model
Conducting Discriminant Analysis
Two-Group Example: Resort Visits
Classification Accuracy and Error Types
Two-Group Example: Telco Churn
Multiple Discriminant Analysis
Stepwise Discriminant Analysis
The Logit Model
Statistical Software
Summary and Exercises

Why Discriminant Analysis?

Real research: Rebate redeemers

A study of 294 consumers looked for the correlates of **rebate proneness**. Predictors: four household shopping-attitude factors plus sex, age and income. The dependent variable had three levels: **nonusers** (no rebate-triggered purchase in 12 months), **light users** (1–2) and **frequent users** (> 2). Multiple discriminant analysis found that (1) the perceived *effort/value relationship* discriminated best – rebate-sensitive consumers see less effort in redeeming and accept smaller refunds; and (2) consumers who know regular prices (and so recognise bargains) respond more to rebates. Dell later used these findings in its Thanksgiving notebook rebates.

Real research: To buy, or not to buy? (Twitter / X, 2022)

During the Twitter takeover, the share of fake accounts was disputed (Twitter: under 5%). Tools such as *Botometer* **score** each account from 0 (human-like) to 5 (bot-like) and then **classify** it (echo-chamber, fake follower, spammer...). Discriminant analysis works the same way: predictors $\rightarrow$ score $\rightarrow$ classification.

Basic Concept of Discriminant Analysis

Discriminant analysis

A technique for analysing data when the **criterion (dependent) variable is categorical** and the **predictor (independent) variables are interval/metric**. Cases are placed into mutually exclusive groups based on a numeric **discriminant score**.

Objectives

1. Develop **discriminant functions**: linear combinations of the predictors that best discriminate between the groups.
2. Examine whether significant differences exist among the groups on the predictors.
3. Determine which predictors contribute most to the intergroup differences.
4. Classify cases into one of the groups based on the predictor values.
5. Evaluate the accuracy of classification.

Two-Group vs. Multiple Discriminant Analysis

- **Two-group discriminant analysis**: the criterion has two categories $\Rightarrow$ only **one** discriminant function.
- **Multiple discriminant analysis**: three or more categories $\Rightarrow$ more than one function may be computed.

How many discriminant functions?

With G groups and k predictors, we can estimate up to $\min(G-1, k)$ functions.

- Defaulters vs. non-defaulters, $k = 5$: $\min(1, 5) = 1$ function.
- Three job types (Headhunters), $k = 3$: $\min(2, 3) = 2$ functions.

The first function has the highest ratio of between- to within-group sum of squares; the second (uncorrelated with the first) the next highest, and so on. Not all functions need be significant.

Applications in Marketing

Typical research questions

- How do store-loyal customers differ demographically from non-loyal ones?
- Do heavy, medium and light soft-drink users differ in frozen-food consumption?
- Which psychographics separate price-sensitive from non-price-sensitive grocery buyers?
- Do market segments differ in media consumption?
- How do patrons of regional vs. national department-store chains differ in lifestyle?
- What distinguishes consumers who respond to direct mail?

Groups and likely predictors

Groups	Predictors
Loan defaulters / non-defaulters	Income, age, household size, outstanding loans
Churners / loyal customers	Tenure, education, expenditure, loyalty-scheme member
Quick upgraders / sclerotic customers	Income, age, past brand switching
Graduate students who complete / withdraw	Exam scores, attendance, financial means

Also: bankruptcy risk from accounting ratios; identifying "influencers"; screening job candidates with aptitude tests.

Relationship to ANOVA and Regression

	ANOVA	Regression	Discriminant / Logit
Similarities			
No. of dependent variables	One	One	One
No. of independent variables	Multiple	Multiple	Multiple
Differences			
Nature of dependent variable	Metric	Metric	Categorical / binary
Nature of independent variables	Categorical	Metric	Metric

- Example: explaining life insurance purchased by age and income. ANOVA: dollars purchased vs. age/income *categories*. Regression: dollars vs. age in years and income in dollars. Discriminant: purchase *category* (high/medium/low) vs. metric age and income.
- With two groups coded 0/1, multiple regression gives coefficients **proportional** to the discriminant coefficients. The binary logit model has the same variable structure as two-group discriminant analysis.

The Discriminant Analysis Model

(Fisher's) linear discriminant function

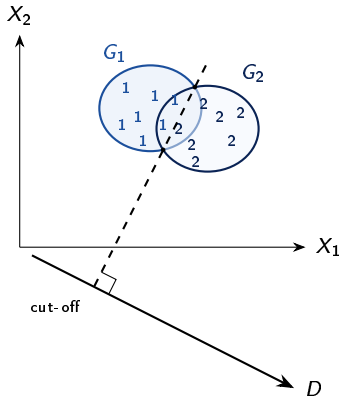
$$D_j = b_0 + b_1 X_{1j} + b_2 X_{2j} + \cdots + b_k X_{kj} = \mathrm{b}' \mathrm{X}_j$$

D_j = discriminant score of case j; b_i = discriminant coefficients (weights); X_i = predictors.

- The weight vector b is **fixed across cases**; only X_j changes from case to case.
- The b's are chosen so the groups differ as much as possible on D: they **maximise the ratio of between-group to within-group sum of squares** of the scores. Any other linear combination gives a smaller ratio.
- Equivalently: maximise homogeneity *within* groups while maximising the distance *between* groups.
- Simplifying assumption: the covariance matrix is the same in every group.

A Geometric Interpretation (Two Groups)

Figure 1. Projecting two groups onto the discriminant axis D



- Two groups, G_1 and G_2, are measured on X_1 and X_2. Each ellipse contains most (say 93%) of its group's points.
- The line through the two intersection points of the ellipses is projected onto a new axis, D.
- On D the univariate distributions overlap **less** than on any other direction – the groups are separated as much as possible.
- Cases are classified by which side of the cut-off their score falls.

Statistics Associated with Discriminant Analysis (1/2)

- **Canonical correlation**: association between the discriminant scores and the groups (between a function and the set of dummies defining group membership).
- **Centroid**: mean discriminant score for a group. One centroid per group; the group means on all functions are the *group centroids*.
- **Classification (confusion / prediction) matrix**: counts of correctly classified (diagonal) and misclassified (off-diagonal) cases. Diagonal sum $\div$ total $=$ **hit ratio**.
- **Discriminant function coefficients** (unstandardised): multipliers of the predictors in their original units.
- **Discriminant scores**: unstandardised coefficients $\times$ predictor values, summed, plus the constant.
- **Eigenvalue**: for each function, the ratio of between- to within-group sum of squares. Large eigenvalues imply superior functions.
- **F values and their significance**: one-way ANOVA with the grouping variable as the factor and each predictor, in turn, as the metric dependent variable.

Statistics Associated with Discriminant Analysis (2/2)

- **Group means and standard deviations**: for each predictor, in each group.
- **Pooled within-group correlation matrix**: obtained by averaging the separate covariance matrices of all groups.
- **Standardised discriminant function coefficients**: multipliers when predictors are standardised to mean 0 and variance 1 – comparable across predictors.
- **Structure correlations** (*discriminant loadings*): simple correlations between each predictor and the discriminant function.
- **Total correlation matrix**: correlations computed treating all cases as one sample.
- **Wilks' λ** (*U statistic*): for each predictor, within-group SS $\div$ total SS. It ranges from 0 to 1. Near 1: group means look the same. Near 0: group means look different.

Assumptions

Each group is a sample from a **multivariate normal** population, and all populations have the **same covariance matrix**.

Another Perspective: Posterior Probabilities

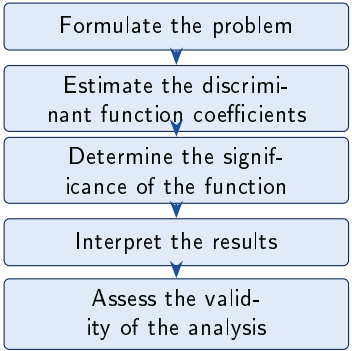
Classification can also be framed as a conditional probability. Given a case's predictors X_j, Bayes' theorem gives

$$P(g_i \mid X_j) = \frac{P(X_j \mid g_i)\, P(g_i)}{\sum_m P(X_j \mid g_m)\, P(g_m)}.$$

- Assign the case to the group with the highest posterior probability (in two groups: to g_i if $P(g_i \mid X_j) > 0.5$).
- Prior probabilities $P(g_i)$: simplest is $1/G$ (SPSS "all groups equal"); use better estimates if they can be justified.
- SPSS saves these posteriors as new variables (e.g. `Dis1_2`, `Dis2_2`), alongside the predicted group (`Dis_1`) and the score (`Dis1_1`). They always sum to 1 across groups because the groups are mutually exclusive and collectively exhaustive.

Conducting Discriminant Analysis

Figure 2. Five steps in conducting discriminant analysis



The steps are illustrated first with **two groups** (vacation-resort visits), then with **three groups** (amount spent on vacation). The same data set is used throughout, so results can be compared.

Additional worked examples:

- **Telco churn** (two groups, $n = 1000$): error types and the confusion matrix.
- **Headhunters** (three job types, $n = 100$): two-function solution.

Step 1: Formulate the Problem

- Identify the **objectives**, the **criterion variable** and the **predictors**.
- The criterion must have two or more **mutually exclusive and collectively exhaustive** categories. A metric criterion must first be categorised (e.g. attitude on a 7-point scale: unfavourable 1–3, neutral 4, favourable 5–7), or cut into equal-size groups.
- Choose predictors from theory or previous research; in exploratory work, the researcher's experience guides selection.

Analysis (estimation) sample and validation (holdout) sample

- The **analysis sample** estimates the discriminant function; the **validation sample** is held back to check it.
- With a large sample, split it in half, estimate on one half, validate on the other, then swap roles: *double cross-validation*.
- Group proportions in each part usually mirror the total sample (e.g. 25% loyal / 75% nonloyal in both).
- Better still, validate repeatedly on different random splits.

Step 2: Estimate the Discriminant Function Coefficients

Two broad approaches:

Direct method

All predictors are included **simultaneously**, regardless of their discriminating power.

Appropriate when theory or previous research says discrimination should be based on all the predictors.

Stepwise discriminant analysis

Predictors are entered **sequentially**, based on their ability to discriminate between the groups.

Appropriate when the researcher wants to select a subset of predictors for the function.

Step 3: Determine the Significance

- H_0: in the population, the means of all discriminant functions are equal in all groups. Interpreting a non-significant function is not meaningful.
- In SPSS the test is based on **Wilks' λ**. With several functions tested together, Wilks' λ is the product of the univariate λ's of each function.
- The p-value comes from a **chi-square transformation** of λ. SAS reports an approximate F instead.
- Functions are tested successively: first all together ("after function 0"), then removing one function at a time.

Reading Wilks' λ

$\lambda = \dfrac{\text{within-group SS}}{\text{total SS}}$: $\lambda \to 1$ means small differences between group mean scores (bad); $\lambda \to 0$ means large differences (good). Reject $H_0 \Rightarrow$ significant discrimination $\Rightarrow$ interpret.

Step 4: Interpret the Results

- Interpretation of the weights is similar to multiple regression: a coefficient's value depends on the other predictors in the function. The **signs are arbitrary**, but show which values lead to large or small scores and hence which group.
- **Do not compare unstandardised coefficients**: predictors are on different scales.
- **Standardised coefficients**: larger absolute values $\Rightarrow$ more discriminating power.
- **Structure correlations**: simple correlations between each predictor and the function, i.e. the variance a predictor shares with the function.
- **Characteristic profile**: describe each group by its means on the most important predictors.

Caveat: multicollinearity

With correlated predictors there is **no unambiguous measure of relative importance**. Standardised coefficients and structure correlations both give "some idea" and must be read with caution – always check the pooled within-groups correlation matrix.

Step 5: Assess Validity

- Multiply the weights estimated on the analysis sample by the predictor values in the **holdout sample**, obtain scores, and assign cases by a decision rule (e.g. nearest centroid).
- Build the classification matrix and compute the **hit ratio** = correctly classified ÷ total.
- **Leave-one-out cross-validation** (LOOCV): the model is re-estimated n times, each time leaving out one case and predicting it. Useful when a large holdout is not possible.
- Classification of the analysis sample itself is **inflated** (it capitalises on chance, i.e. overfitting): expect it to beat LOOCV and holdout results.

Compare with chance

With equal group sizes, chance accuracy $= 1/G$. Rule of thumb: classification accuracy should be at least **25% greater than chance** (e.g. two equal groups: chance 50%, so aim for at least 62.5%).

Example: Who Visits a Vacation Resort?

- Goal: the salient characteristics of families that visited a resort in the last two years.
- Pretest of 42 households: **30 in the analysis sample**, **12 in the holdout**, both balanced on Visit (15/15 and 6/6).
- Criterion: **Visit** = 1 (visited), 2 (did not visit).
- Predictors: annual family **Income** ($000), attitude toward **Travel** (9-point), importance of family **Vacation** (9-point), household size (**Hsize**), **Age** of head of household.
- A sixth variable, **Amount** spent on vacation (low/medium/high), is used later for the three-group example.

Group means	Income	Travel	Vacation	Hsize	Age
Visit = 1	60.52	5.40	5.80	4.33	53.73
Visit = 2	41.91	4.33	4.07	2.80	50.13
Std. dev. (group 1 / group 2)	9.83 / 7.55	1.92 / 1.95	1.82 / 2.05	1.23 / 0.94	8.77 / 8.27

Resort Data: Analysis Sample ($n = 30$)

No.	Visit	Income	Travel	Vacation	Hsize	Age	Amount
1	1	50.2	5	8	3	43	M (2)
2	1	70.3	6	7	4	61	H (3)
3	1	62.9	7	5	6	52	H (3)
4	1	48.5	7	5	5	36	L (1)
5	1	52.7	6	6	4	55	H (3)
6	1	75.0	8	7	5	68	H (3)
7	1	46.2	5	3	3	62	M (2)
8	1	57.0	2	4	6	51	M (2)
9	1	64.1	7	5	4	57	H (3)
10	1	68.1	7	6	5	45	H (3)
11	1	73.4	6	7	5	44	H (3)
12	1	71.9	5	8	4	64	H (3)
13	1	56.2	1	8	6	54	M (2)
14	1	49.3	4	2	3	56	H (3)
15	1	62.0	5	6	2	58	H (3)

No.	Visit	Income	Travel	Vacation	Hsize	Age	Amount
16	2	32.1	5	4	3	58	L (1)
17	2	36.2	4	3	2	55	L (1)
18	2	43.2	2	5	2	57	M (2)
19	2	50.4	5	2	4	37	M (2)
20	2	44.1	6	6	3	42	M (2)
21	2	38.3	6	6	2	45	L (1)
22	2	55.0	1	2	2	57	M (2)
23	2	46.1	3	5	3	51	L (1)
24	2	35.0	6	4	5	64	L (1)
25	2	37.3	2	7	4	54	L (1)
26	2	41.8	5	1	3	56	M (2)
27	2	57.0	8	3	2	36	M (2)
28	2	33.4	6	8	2	50	L (1)
29	2	37.5	3	2	3	48	L (1)
30	2	41.3	3	3	2	42	L (1)

Holdout sample: 12 further households (6 visitors, 6 non-visitors) on the same variables.

Two-Group Results: Univariate Tests and the Function

Univariate tests (F with 1 and 28 df)

	Wilks' λ	F	Sig.
Income	0.453	33.80	0.000
Travel	0.925	2.28	0.143
Vacation	0.824	5.99	0.021
Hsize	0.657	14.64	0.001
Age	0.954	1.34	0.257

Individually, only **Income**, **Vacation** and **Hsize** differ significantly. Pooled within-group correlations are all low ($|r| \leq 0.20$): multicollinearity is unlikely to be a problem.

Canonical discriminant function

Eigenvalue	1.7862 (100% of variance)
Canonical correlation	0.8007
Wilks' λ (after function 0)	0.3589
Chi-square (df $= 5$)	26.130, Sig. 0.0001

- Two groups $\Rightarrow$ one function.
- $0.8007^2 = 0.64$: 64% of the variance in Visit is accounted for by the model.
- $\lambda = 0.3589$ is significant beyond the 0.05 level, so we interpret.

Two-Group Results: Relative Importance

	Standardised coefficient	Structure correlation	Unstandardised coefficient
Income	0.743	0.822	0.0848
Hsize	0.469	0.541	0.4274
Vacation	0.233	0.346	0.1203
Travel	0.096	0.213	0.0496
Age	0.209	0.164	0.0245
(constant)			−7.9755

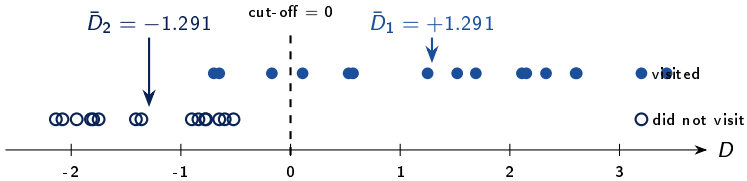
Estimated function

$$D = -7.975 + 0.085\,\text{Income} + 0.050\,\text{Travel} + 0.120\,\text{Vacation} + 0.427\,\text{Hsize} + 0.025\,\text{Age}$$

- Both standardised coefficients and structure correlations rank **income** first, then **household size**, then **importance of vacation**.
- All coefficients are positive: higher income, household size, vacation importance, travel attitude and age make a visit more likely.
- **Characteristic profile**: visitors have about \$60.5k income vs. \$41.9k, 4.3 vs. 2.8 members, vacation importance 5.8 vs. 4.1.

Two-Group Results: Centroids and Scores

Figure 3. Discriminant scores of the 30 analysis-sample households (computed)



Equal group sizes $\Rightarrow$ cut-off $= (1.291 + (-1.291))/2 = 0$. Positive score $\Rightarrow$ classify as visitor. Three visitors (households 1, 7 and 14) fall below 0 and are misclassified; no non-visitor scores above 0.

Two-Group Results: Validation

	Actual	Predicted	
		Visit 1	Visit 2
Original	1	12	3
	2	0	15
Cross-validated	1	11	4
(LOOCV)	2	2	13
Holdout ($n = 12$)	1	4	2
	2	0	6

Hit ratios

- Analysis sample: $(12 + 15)/30 = 90.0\%$ – likely inflated, as the same data estimated the model.
- LOOCV: $(11 + 13)/30 = 80.0\%$.
- Holdout: $(4 + 6)/12 = 83.3\%$.
- Chance (two equal groups): $1/2 = 50\%$.

Verdict

Improvement over chance is well over 25% $\Rightarrow$ validity is **satisfactory**.

Real Research: Satisfied Salespeople Stay

Real research: Stayers vs. leavers

In a slow-growth economy, 85% of businesspeople worried about recruiting and 81% about retaining employees. Discriminant analysis compared salespeople who **left** a large computer manufacturer with those who **stayed**.

Predictors (26): company rating, job security, 7 job-satisfaction, 4 role-conflict, 4 role-ambiguity and 9 sales-performance measures.

Canonical correlation $R = 0.4572$; Wilks' $\lambda = 0.7909$, $F(26, 173) = 1.7588$, $p = 0.018$.

Stayers found the job more exciting, satisfying, challenging and interesting.

Top predictors	Std. coef.	Structure r
1. Work (satisfaction)	0.391	0.545
2. Promotion (satisfaction)	0.152	0.504
3. Job security	0.138	0.496
4. Customer relations (perf.)	0.175	0.491
5. Company rating	0.324	0.482

Job satisfaction and opportunity for promotion are the top discriminators by structure correlation, followed by job security.

Anomaly: promotion is 2nd by structure correlation but *not* by standardised coefficient. This results from multicollinearity.

Real Research: Home Bodies and Couch Potatoes

Real research: TV users vs. nonusers among the elderly

370 elderly consumers (185 users, 185 nonusers) were scored on five factor dimensions. The function was estimated on 142 respondents and cross-validated on 198 (30 dropped for missing values).

Canonical correlation 0.4291 ($p < 0.0001$); eigenvalue 0.2257.

Home orientation contributed most; morale, security and health, and respect also contributed; the social factor added little.

Analysis-sample hit ratio 68.3%; cross-validation 75.8%. The same approach could target Generation X users of interactive TV.

Dimension	Std. coefficient
Home orientation	0.775
Security and health	0.399
Respect	0.321
Morale	0.278
Social	−0.020

Cross-validation	Nonusers	Users
TV nonusers (108)	85	23
TV users (90)	25	65

$(85 + 65)/198 = 75.8\%$ vs. roughly 50% by chance.

The Confusion Matrix and Asymmetric Errors

Loan applicants (positive = non-defaulter, advance the loan)

Actual \ Pred.	Defaulter (refuse)	Non-defaulter (lend)
Defaulter	True negative	**False positive**
Non-defaulter	**False negative**	True positive

- **False positive**: lend to a defaulter $\Rightarrow$ a realised bad-debt loss.
- **False negative**: refuse a good borrower $\Rightarrow$ forgone profit, and possibly the customer's whole lifetime value.

Key takeaway

Misclassifications can have **asymmetric consequences**. Look beyond overall accuracy to the rates of false positives and false negatives, and decide which error is worse *in this setting*. That is a judgement call.

The same holds with three ordered groups: predicting a *high* spender as *low* is a worse error than predicting *medium*, so two models with the same hit ratio need not be equally good.

Performance Measures from the Confusion Matrix

Actual	Neg.	Pos.	Total
Negative	n_{11}	n_{12}	$n_{1\cdot}$
Positive	n_{21}	n_{22}	$n_{2\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 2}$	n

n_{11} TN, n_{12} FP, n_{21} FN, n_{22} TP.

Measure	Formula	Want
Overall predictive accuracy	$\text{OPA} = (n_{11} + n_{22})/n$	large
True positive rate	$\text{TPR} = n_{22}/n_{2\cdot}$	large
True negative rate	$\text{TNR} = n_{11}/n_{1\cdot}$	large
False positive rate*	$\text{FPR} = n_{12}/n_{\cdot 2}$	small
False negative rate*	$\text{FNR} = n_{21}/n_{\cdot 1}$	small

*** A note on definitions**

The source text divides FPR and FNR by the *predicted* totals, i.e. the share of positive (negative) predictions that are wrong. Many texts call these the *false discovery* and *false omission* rates, and instead define $\text{FPR} = n_{12}/n_{1\cdot} = 1 - \text{TNR}$ and $\text{FNR} = n_{21}/n_{2\cdot} = 1 - \text{TPR}$. State which convention you use.

Training Data, Test Data and Overfitting

- **Training data** estimate the parameters; **test data** assess out-of-sample performance.
- **LOOCV**: each case takes a turn as the test case while the other $n - 1$ cases form the training data.
- If all cases are used for training, accuracy measures suffer from **overfitting**: the model is credited for correctly classifying the very cases used to estimate it.
- So compute OPA (and the other measures) on the **cross-validated** classification: lower, but more realistic.

How large was the overfitting gap in our examples?

Resort (two groups): 90.0% → 80.0% Telco churn: 65.6% → 65.5% Headhunters: 78.0% → 74.0%
Resort (three groups): 86.7% → 66.7%

Example: Predicting Customer Churn

`telco.sav` (ships with SPSS): $n = 1000$ customers, groups G_1 = no churn (726), G_2 = churned in the last month (274). Five predictors, so one function.

	No churn		Churn		Univariate test	
Predictor	Mean	SD	Mean	SD	F	Sig.
X_1 Months with service	40.47	20.57	22.43	17.59	165.17	$< .001$
X_2 Age in years	43.63	12.72	36.52	10.52	68.15	$< .001$
X_3 Years at current address	13.06	10.55	7.56	7.39	62.67	$< .001$
X_4 Household income ($000)	83.54	119.40	61.63	60.97	8.40	.004
X_5 People in household	2.32	1.43	2.35	1.45	0.10	.756

- "Good" predictors have very different group means and small SDs: churners had been with the firm about 22 months vs. 40 for stayers.
- Household size barely differs ($p = .756$): a candidate for removal.
- Wilks' $\lambda = 0.851$, $\chi^2_5 = 161.09$, $p < .001$: the function discriminates significantly.

Churn: Estimated Function and Importance

	Unstd.	Std.	Structure
Months with service	0.043	0.847	0.971
Age in years	0.021	0.251	0.623
Years at address	0.006	0.059	0.598
Household income	−0.001	−0.057	0.219
People in household	0.055	0.079	−0.024
(constant)	−2.537		

$D_j = -2.537 + 0.043X_1 + 0.021X_2 + 0.006X_3 - 0.001X_4 + 0.055X_5$

Ranges differ hugely (income 9 to 1668; household 1 to 8), so compare **standardised** coefficients, not raw ones. Largest within-group correlation: age and address, $r = 0.638$, so multicollinearity is mild.

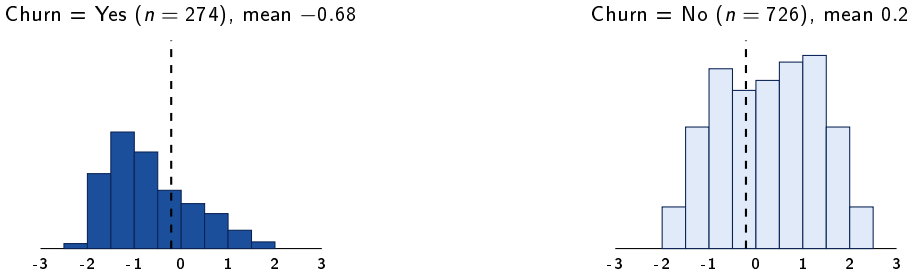
Centroids: no churn $= 0.257$, churn $= -0.682$
Cut-off $c = (0.257 - 0.682)/2 = -0.2125$

Customer 1: tenure 13, age 44, address 9, income 64, household 2:
$D_1 = -2.537 + 0.043(13) + 0.021(44) + 0.006(9) - 0.001(64) + 0.055(2) \approx -0.94$
$-0.94 < -0.2125 \Rightarrow$ classified as **churner** (correct).

Signs (low score = churn), ceteris paribus: longer tenure, older, longer at address and larger household $\Rightarrow$ less likely to churn; higher income $\Rightarrow$ more likely to churn.

Churn: Distribution of Discriminant Scores

Figure 4. Discriminant scores by group (bar heights approximate, read from the source histograms)



Churners have low, right-skewed scores; non-churners are spread symmetrically around 0.26. Many non-churners fall left of the cut-off (-0.2125, dashed), so expect many **false positives** (stayers flagged as churners).

Churn: Confusion Matrix and Error Rates

	Actual	Pred. No	Pred. Yes
Original	No	458	268
	Yes	76	198
Cross-val.	No	457	269
	Yes	76	198

Cross-validated (positive = churn):
OPA $= (457 + 198)/1000 = 65.5\%$ (original 65.6%)
TPR $= 198/274 = 72.3\%$; TNR $= 457/726 = 62.9\%$
FPR $= 269/467 = 57.6\%$; FNR $= 76/533 = 14.3\%$
(source convention: share of positive / negative predictions that are wrong)

- **False positive**: we expect churn but the customer stays. A retention discount is wasted on someone who would have stayed – 57.6% of churn predictions are like this.
- **False negative**: we expect no churn but the customer leaves. No intervention, so the customer's lifetime value is lost – only 14.3% of no-churn predictions.
- Correct about two-thirds of the time. Whether that is good enough depends on how costly each error is: *your* judgement call.

Managerial implication

Tenure and age matter most: new, young customers are the high-risk churners, so focus retention effort on them.

Three-Group Example: Amount Spent on Vacation

Same 30 households, now grouped by **Amount** spent on family vacation: low (1), medium (2), high (3) – ten each. Same five predictors.

Group means	Income	Travel	Vacation	Hsize	Age
Low (1)	38.57	4.50	4.70	3.10	50.30
Medium (2)	50.11	4.00	4.20	3.40	49.50
High (3)	64.97	6.10	5.90	4.20	56.00
Wilks' λ	0.262	0.788	0.881	0.874	0.882
F (2, 27 df)	38.00	3.63	1.83	1.94	1.80
Sig.	0.000	0.040	0.180	0.163	0.184

- Income separates the groups most widely; there is some separation on travel and vacation.
- Groups 1 and 2 are very close on household size and age; age has a large SD relative to the separation.
- Some within-group correlation of vacation and household size with income, and of age with travel – worth watching, but probably not serious.
- Individually, only **income** and **travel** are significant.

Three Groups: Estimation and Significance

$G = 3$ groups, $k = 5$ predictors $\Rightarrow$ up to $\min(2, 5) = 2$ functions.

Fn	Eigenvalue	% var.	Cum. %	Can. corr.
1	3.8190	93.93	93.93	0.8902
2	0.2469	6.07	100.00	0.4450

After fn	Wilks' λ	χ^2	df	Sig.
0	0.1664	44.831	10	0.00
1	0.8020	5.517	4	0.24

- Function 1 has a large eigenvalue and explains almost 94% of the explained variance – it is likely to be superior.
- Testing **both** functions together (after 0): $\lambda = 0.1664$, significant – jointly they discriminate among the three groups.
- With function 1 removed, $\lambda = 0.8020$ is **not** significant: function 2 adds nothing significant.

Three Groups: Interpreting the Functions

	Standardised		Structure	
	Fn 1	Fn 2	Fn 1	Fn 2
Income	**1.047**	−0.421	**0.856***	−0.278
Hsize	−0.163	0.129	**0.193***	0.077
Vacation	−0.142	**0.534**	0.149	**0.454***
Travel	0.340	**0.769**	0.219	**0.588***
Age	0.495	**0.524**	0.166	**0.341***

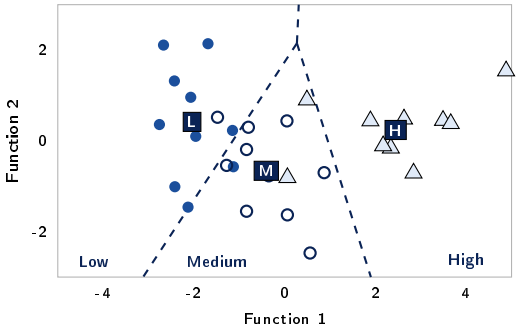
* largest absolute correlation of each variable with any function.

Centroids (Fn 1, Fn 2): Low (−2.041, 0.418); Medium (−0.405, −0.659); High (2.446, 0.240).

- **Function 1** ≈ income and household size: it orders the groups high > medium > low. Higher income and larger households spend more.
- **Function 2** ≈ travel, vacation and age: it separates low (highest) from medium (lowest).
- So low spenders like travel and value vacations *more* than medium spenders. Why spend less? Perhaps they would like to but cannot afford it.

Three Groups: All-Groups Scattergram and Territorial Map

Figure 5. Case scores, group centroids and territorial boundaries (computed)



- • low, ○ medium, △ high; squares = centroids (L, M, H).
- Group 3 sits highest on function 1 and group 1 lowest; function 2 separates low (up) from medium (down).
- **Territorial map**: each case is assigned to the nearest centroid, so the dashed lines are the boundaries between group territories.
- Points on the wrong side of a boundary are misclassifications (e.g. two high spenders fall in the medium territory).

Three Groups: Validation

	Actual	Pred. L	Pred. M	Pred. H
Original	Low	9	1	0
	Medium	1	9	0
	High	0	2	8
LOOCV	Low	7	3	0
	Medium	4	5	1
	High	0	2	8
Holdout	Low	3	1	0
	Medium	0	3	1
	High	1	0	3

- Analysis sample: $(9 + 9 + 8)/30 = 86.7\%$.
- LOOCV: $(7 + 5 + 8)/30 = 66.7\%$.
- Holdout: $(3 + 3 + 3)/12 = 75.0\%$.
- Chance with three equal groups: $1/3 = 33.3\%$, so the improvement exceeds 25%: at least satisfactory validity.

Not all errors are equal

In the holdout, one **high** spender was predicted **low** – a worse error than predicting medium. Two models with the same hit ratio can differ in the severity of their mistakes.

Real Research: The Home Is Where the Patient's Heart Is

Real research: Four health care delivery systems

102 consumers rated home health care, hospitals, nursing homes and outpatient clinics on 10 attributes. Multiple discriminant analysis found **three** functions, all significant at 0.01 (63.0%, 29.4% and 7.6% of discriminating power).

- Fn 1 **personalised care**: comfort (0.53), privacy (0.40).
- Fn 2 **quality of medical care**: best care (0.67), faster recovery (0.32).
- Fn 3 **value**: sanitation (−0.70), expense (0.52).

Correct classification ranged from 86% (hospitals) to 68% (outpatient clinics).

Centroids	Fn 1	Fn 2	Fn 3
Hospital	−1.66	0.97	−0.08
Nursing home	−0.60	−1.36	−0.27
Outpatient clinic	0.54	−0.13	0.77
Home health care	1.77	0.50	−0.39

- Home health care is most favourable on personalised care; hospitals least.
- Nursing homes are well separated from the rest on quality of medical care.
- Outpatient clinics offer the best value.

Example: Headhunters – Matching Personality to Job Type

A recruitment agency tested 100 previously placed candidates in three role types: (1) front-of-house, (2) back office, (3) remote. Predictors: interest in outdoor activities (X_1), attitude toward social events (X_2), reserved score (X_3). $\min(3-1, 3) = 2$ functions.

Means	Outdoor	Social	Reserved
Front-of-house (33)	11.61	25.64	9.27
Back office (33)	18.00	20.76	10.03
Remote (34)	15.44	15.32	13.35
Univariate F (2, 97)	17.21	48.03	14.85

All three $p < .001$; predictor intercorrelations ≤ 0.08, so no multicollinearity concern.

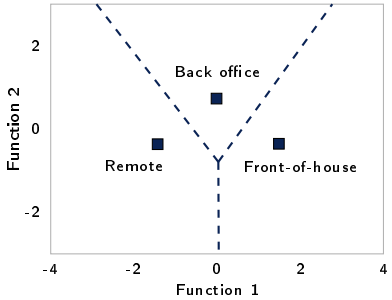
$$D_1 = -1.494 - 0.064X_1 + 0.200X_2 - 0.150X_3$$
$$D_2 = -2.319 + 0.206X_1 + 0.028X_2 - 0.125X_3$$

Test	λ	χ^2	df	Sig.
1 through 2	.319	109.74	6	$< .001$
2	.787	23.05	2	$< .001$

Both functions significant. Standardised coefficients on Fn 1: social 0.860, reserved -0.489, outdoor -0.287 – social attitude is about 1.76 times as important as reserved score and 3 times as important as outdoor interest.

Headhunters: Centroids, Territories and Accuracy

Figure 6. Headhunters group centroids and territorial boundaries in discriminant space



Centroids: front $(1.486, -0.355)$, back $(-0.013, 0.731)$, remote $(-1.429, -0.365)$. High $D_1 \Rightarrow$ front-of-house (sociable); low $D_1 \Rightarrow$ remote (reserved, outdoor-oriented).

LOOCV	Front	Back	Remote
Front-of-house	23	9	1
Back office	7	22	4
Remote	0	5	29

OPA: original 78/100; cross-validated $(23 + 22 + 29)/100 = 74\%$ vs. chance $\approx 33\%$. No remote case was classified front-of-house.

With three groups, true/false positive rates are not defined. Instead weigh severity: an extreme introvert placed front-of-house (anxiety) may be worse than an extrovert placed remotely (boredom).

Stepwise Discriminant Analysis

- Analogous to stepwise regression: predictors are entered **sequentially** by their ability to discriminate.
- An F ratio is computed for each predictor (univariate ANOVA: groups as factor, predictor as criterion). The one with the highest F enters first, if it meets significance and tolerance criteria.
- The next predictor enters on the highest **partial** F, given those already in. Each predictor in the function is re-tested for retention.
- Continue until no remaining predictor meets the criteria. A summary of variables entered and removed is reported, plus the usual direct-method output.

The Mahalanobis procedure

A stepwise criterion that maximises a generalised measure of the distance between the two **closest** groups, making maximal use of the available information.

Resort example (two groups): Income entered first, then household size, then vacation importance – the same order of importance found by the direct method's standardised coefficients and structure correlations.

The Binary Logit Model

With a binary dependent variable (coded 0/1) and metric predictors we can use two-group discriminant analysis, OLS regression, logit or probit. Discriminant analysis asks *which group* a case belongs to; logit estimates the *probability* of belonging – it sits between regression and discriminant analysis.

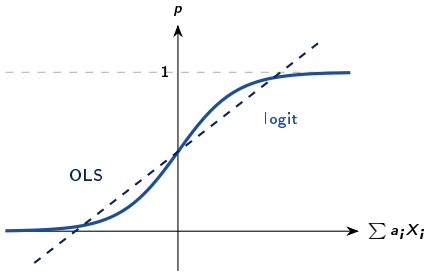
Binary logit (logistic regression)

$$\log_e\left(\frac{p}{1-p}\right) = a_0 + a_1X_1 + \cdots + a_kX_k$$

$$p = \frac{\exp\left(\sum_{i=0}^{k} a_iX_i\right)}{1 + \exp\left(\sum_{i=0}^{k} a_iX_i\right)}$$

p = probability of success; a_i = parameters to estimate.

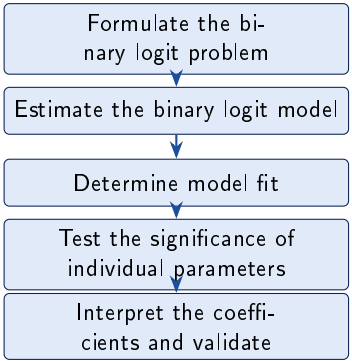
Figure 7. Logit vs. linear probability



As $X \to -\infty$, $p \to 0$; as $X \to +\infty$, $p \to 1$. OLS ($p = \sum a_iX_i$) is not constrained to $[0, 1]$.

Conducting Binary Logit Analysis

Figure 8. Five steps in conducting binary logit analysis



- **Formulate**: state objectives; identify the binary criterion and the predictors. Categorical predictors can enter as dummy variables. Split into analysis and validation samples if possible.
- **Estimate**: OLS minimises squared errors, but in logit each error takes only two values ($-p$ if $Y = 0$, $1 - p$ if $Y = 1$). Parameters are estimated by **maximum likelihood**: those that make the observed data most likely.
- **Fit**: likelihood-based pseudo-R^2 and the classification table.
- **Significance**: Wald statistic.
- **Interpret and validate**: log-odds, odds ratios, holdout hit ratio.

Logit: Model Fit, Significance and Interpretation

- **Model fit**: -2 log likelihood; **Cox & Snell** R^2 and **Nagelkerke** R^2 play the role of R^2. Cox & Snell cannot reach 1.0 even with a perfect fit; Nagelkerke rescales it so it can.
- **Classification**: predict $Y = 1$ if $\hat{p} > 0.5$, else $Y = 0$; compare predicted with actual for the percentage correct.
- **Wald statistic**: Wald $= (a_i / SE_{a_i})^2$, chi-square with 1 df for a metric predictor (categories -1 for a nonmetric one). For practical purposes a t (or, with many observations, z) test on a_i / SE_{a_i} is equivalent.
- **Interpretation**: a one-unit increase in X_i raises the **log odds** by a_i, holding the other predictors constant. The sign tells whether the probability rises or falls; e^{a_i} is the **odds ratio**.
- Validation mirrors discriminant analysis: estimate on the analysis sample, classify the validation sample, compute the hit ratio.

Illustration: Explaining Brand Loyalty

30 respondents: 15 brand loyal (1), 15 not (0). Predictors, all on 1–7 scales: attitude toward the **Brand**, the **Product** category, and **Shopping**.

OLS first (to show its limitation):
$\hat{p} = -0.684 + 0.183\,\text{Brand} + 0.020\,\text{Product} + 0.074\,\text{Shopping}$

Only the constant and Brand are significant at 0.05.

- Brand = Product = Shopping = 1: $\hat{p} = -0.684 + 0.277 = -0.41$ (negative!)
- All = 7: $\hat{p} = -0.684 + 1.939 = 1.26$ (> 1!)

A probability outside $[0, 1]$ is conceptually unappealing; logistic regression avoids it.

Logit results

	B	S.E.	Wald	Sig.	Exp(B)
Brand	1.274	.479	7.075	.008	3.575
Product	.186	.322	.335	.563	1.205
Shopping	.590	.491	1.442	.230	1.804
Constant	−8.642	3.346	6.672	.010	.000

$-2\text{LL} = 23.471$; Cox & Snell $R^2 = .453$; Nagelkerke $R^2 = .604$ – a reasonable fit.

Classification (cut 0.5): 12 of 15 non-loyal and 12 of 15 loyal correct, so **80%** overall.

Brand Loyalty: Interpretation

- Only **attitude toward the brand** is significant: Wald $= (1.274/0.479)^2 = 7.08$, $p = .008$.
- Its positive sign: a more positive brand attitude means higher loyalty. Each extra scale point raises the log odds of loyalty by 1.274, i.e. multiplies the odds by $e^{1.274} \approx 3.6$.
- Attitudes toward the product category and toward shopping do not influence loyalty.
- **Managerial implication**: to increase loyalty, foster a more positive attitude toward the brand – not toward the category or shopping.
- Unlike discriminant analysis, logit provides **standard errors**, so the significance of each coefficient can be assessed.
- With more than two categories, the model generalises to the **multinomial logit**.

Real Research: Satisfaction Programs and Ethical Firms

Real research: Satisfactory results of satisfaction programs in Europe

HP found that European customers over 40 struggled with technical aspects and valued customer service programs, while younger customers valued technical features. A two-group discriminant analysis (satisfied vs. dissatisfied) confirmed that the variety and scope of customer service programs strongly differentiated the groups. HP launched feedback, satisfaction-survey and total-quality programs and later a "total customer experience and quality" framework.

Real research: Discriminant analysis discriminates ethical and unethical firms

149 firms reported how they behave in 18 ethical situations (9 marketing-related: misleading sales presentations, gifts for preferential treatment, pricing below out-of-pocket costs...) and were split into "never practice" vs. "practice". Two-group discriminant analysis showed that **attitudes** and **company size** best predicted ethical behaviour – smaller firms tended to be more ethical on marketing issues.

Running the Analyses in SPSS

Discriminant: `Analyze > Classify > Discriminant`

1. Move the criterion (e.g. Visit) to `Grouping Variable`; `Define Range` (1 to 2; 1 to 3 for Amount).
2. Move predictors to `Independents`; choose `Enter independents together` (or `Use stepwise method`).
3. `Statistics`: Means, Univariate ANOVAs, Within-groups correlation, Unstandardized coefficients.
4. `Classify`: priors All groups equal; Summary table; Leave-one-out classification; Within-groups covariance.
5. `Save`: predicted group, discriminant scores, probabilities of membership.
6. To validate on a holdout, use a selection variable for the analysis cases.

Logit: `Analyze > Regression > Binary Logistic`

1. Move Loyalty to `Dependent`.
2. Move Brand, Product, Shopping to `Covariates`.
3. Method: `Enter`. Click OK.

SAS Enterprise Guide

- `Analyze > Multivariate > Discriminant Analysis`: classification variable + analysis variables; request univariate tests, summary statistics and cross-validation (`CAN`, `PCORE`).
- `Analyze > Regression > Logistic Regression`: main effects; show classification table with a 0.5 cut-off.

MINITAB: `Stat > Multivariate > Discriminant Analysis`. Not available in Excel.

Summary

- Discriminant analysis: **categorical criterion**, **metric predictors**. Two groups $\Rightarrow$ one function; G groups and k predictors $\Rightarrow$ up to $\min(G-1, k)$.
- Five steps: formulate (incl. analysis/holdout split), estimate (direct or stepwise), test significance (Wilks' λ, chi-square), interpret (standardised coefficients, structure correlations, profiles), validate (classification matrix, hit ratio vs. chance).
- Relative importance is never unambiguous under multicollinearity.
- Validate out of sample or with LOOCV – analysis-sample accuracy is optimistic – and weigh the **cost of each error type**, not just overall accuracy.
- Logit (logistic regression) is an alternative when the criterion is binary: it estimates the probability of a binary event, constrained to $[0, 1]$ (unlike OLS), and gives standard errors (unlike discriminant analysis).

Remember: DISCRIMINANT

The steps and key concepts as an acronym

- **D**ependent variable: categorical
- **I**ndependent variables: metric
- **S**tructure correlations or discriminant loadings
- **C**alculation of the discriminant function
- **R**elative importance of predictors: ambiguous
- **I**nterpretation: scattergram and territorial map
- **M**eans and standard deviations for groups
- **I**nference: determination of significance
- **N**umber of functions possible: $\min(G-1, k)$
- **A**ssociation: canonical correlation
- **N**umber 1 function has the highest eigenvalue
- **T**esting for validity: classification analysis

Worked Problem: Computing Discriminant Scores

Given the unstandardised coefficients below and a constant of 2.04, compute each respondent's score.

ID	Age	Income	Risk taking	Optimistic
coefficient	0.38	0.44	−0.39	1.26
0246	36	43.7	21	65
1337	44	62.5	28	56
2375	57	33.5	25	40
2454	63	38.7	16	36

$D = 2.04 + 0.38\,\text{Age} + 0.44\,\text{Income} - 0.39\,\text{Risk} + 1.26\,\text{Optimistic}$

Respondent 0246:
$2.04 + 13.68 + 19.228 - 8.19 + 81.90 = 108.658$

ID	Score
0246	108.658
1337	105.900
2375	79.090
2454	82.128

Exercises and Discussion

Is convenience orientation more important than income?

Heavy, light and non-users of frozen foods: the two largest standardised coefficients are 0.97 (convenience orientation) and 0.61 (income). **Answer**: not unequivocally – without the correlation between the predictors, multicollinearity may be present; still, given the magnitudes, the conclusion seems reasonable.

An airline researcher's assumption

The figure for flight-time convenience was 0.80 and for price 0.65. She concludes that customers care more about convenient flight times than price. Is she correct? Discuss what such coefficients do and do not tell her (scale, correlation between predictors).

Exercises and Discussion (continued)

Benetton usage groups (data file `Benetton.sav`)

45 participants (5 with missing values) classified as light, medium or heavy users; predictors: awareness, attitude, preference, intention, loyalty (7-point). Do the groups differ when the predictors are considered simultaneously? *Key results*: Fn 1 explains 96.7% of the explained variance; both functions jointly: $\lambda = .135$, $\chi^2_{10} = 70.151$, $p < .001$; Fn 2 alone: $\lambda = .846$, $\chi^2_4 = 5.858$, $p = .210$ (not significant); only awareness, attitude and preference differ univariately; LOOCV hit ratio 80%.

Discussion

Is it meaningful to determine the relative importance of predictors in discriminating between the groups? Why or why not? Is it meaningful to use eigenvalues to express the ratio of explained to unexplained variation?

Review Questions

1. What are the objectives of discriminant analysis?
2. Describe the relationship of discriminant analysis to regression and ANOVA.
3. What are the steps in conducting discriminant analysis?
4. What conditions must the criterion variable meet?
5. How should the sample be split for estimation and validation?
6. What is Wilks' λ? For what is it used?
7. Define discriminant scores and eigenvalues.
8. What is a classification matrix? What is the hit ratio?
9. Explain structure correlations.
10. How would you interpret discriminant weights?
11. How is the validity of discriminant analysis assessed?
12. With equal group sizes, how is chance accuracy found?
13. How does the stepwise procedure differ from the direct method?
14. In a binary setting, what are the four possible classification outcomes? Which error is worse – and what does it depend on?
15. Can a case belong to more than one group? Why or why not?
16. Which predictors would you use to discriminate loan defaulters from non-defaulters?
17. What is a "good" number of predictors?
18. Why is logit preferred to OLS for a binary criterion, and what does logit offer that discriminant analysis does not?